WORLD HEALTH ORGANIZATION (WHO)

Guideline for the Pharmacological Treatment of Hypertension in Adults

Official WHO Clinical Practice Guideline (Geneva, 2021) - Recommendations

Official Reference: WHO Guidelines Approved by the Guidelines Review Committee (ISBN 978-92-4-003398-6)
Official Web Portal: https://www.who.int/publications/i/item/9789240033986

1. Blood Pressure Thresholds for Pharmacological Initiation

* Initiate pharmacological treatment in any individual with confirmed SBP >=140 mmHg or DBP >=90 mmHg.
* Initiate treatment in individuals with existing CVD and SBP 130-139 mmHg without delay.
* Initiate treatment in individuals without CVD but with high cardiovascular risk, diabetes, or CKD and SBP
* Confirmation requires repeated clinic measurements over 2-3 visits or verified ambulatory/home monitoring.

2. Recommended First-Line Antihypertensive Drug Classes

* WHO recommends any of the following 3 first-line drug classes for initial therapy:

1. Thiazide or thiazide-like diuretics (Hydrochlorothiazide, Chlorthalidone, Indapamide).
2. Long-acting dihydropyridine Calcium Channel Blockers (CCBs - e.g. Amlodipine).
3. Angiotensin-Converting Enzyme Inhibitors (ACE-I - e.g. Enalapril) OR Angiotensin Receptor Blockers (ARBs

* Strict Clinical Safety Note: Never combine an ACE-I with an ARB concurrently due to nephrotoxicity.
* Beta-blockers are not recommended as first-line unless there is a specific compelling cardiac indication.

3. Dual Combination Therapy Strategy

* Combination therapy with 2 first-line agents from different classes is recommended for initial therapy.
* Especially indicated when baseline blood pressure is >=20 mmHg systolic or >=10 mmHg diastolic above targe
* Single-pill fixed-dose combinations (FDCs) improve medication adherence and blood pressure control rates.

4. Target Blood Pressure Goals & Follow-Up Protocols

* Universal Target: Blood pressure <140/90 mmHg for all adult hypertension patients without co-morbidities.
* Strict Target: Blood pressure <130/80 mmHg for patients with co-morbid diabetes mellitus, CKD, or known CV
* Follow-up interval: Monthly visits until blood pressure reaches target; every 3-6 months once stabilized.

5. Baseline Clinical & Laboratory Assessment Package

* Assessment of serum creatinine and estimated glomerular filtration rate (eGFR).
* Assessment of serum potassium and sodium electrolytes (prior to and during diuretic/ACE-I/ARB therapy).
* Fasting lipid profile, fasting blood glucose or HbA1c.
* Complete urine analysis (dipstick for protein/microalbuminuria).
* 12-lead Electrocardiogram (ECG) for baseline cardiac rhythm and LVH evaluation.